



THE UNIVERSITY OF
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FINAL REPORT

The Art of Scientific Computation Power Laws and Finite Size Scaling in the BTW Abelian Sandpiles Model

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Contents

1	Introduction	2
2	Simulation	3
3	Power Law Fit	4
3.1	Naïve Power Law Fit	4
3.2	Manually-truncated Power Law	7
4	Finite Size Scaling Effect	9
4.1	Exponentially Truncated Power Law	9
5	Non-FSS Behaviour of Toppling Number	12
	Bibliography	12

Chapter 1

Introduction

1. Introduce the BTW sandpiles model as a model for self-organised criticality and $1/f$ noise [1]
2. Define criticality, self-organised criticality, power law scaling, etc
3. Mathematically describe the BTW model and toppling condition
4. Describe modelling choices for power law and truncated power law distributions
5. Justify the validity of distribution modelling using markov chain theory

Chapter 2

Simulation

- Describe the simulation procedure including implementation details (numba, burn-in)
- Show plots of basic results found
- Describe the qualitative scaling behaviour observed across different L

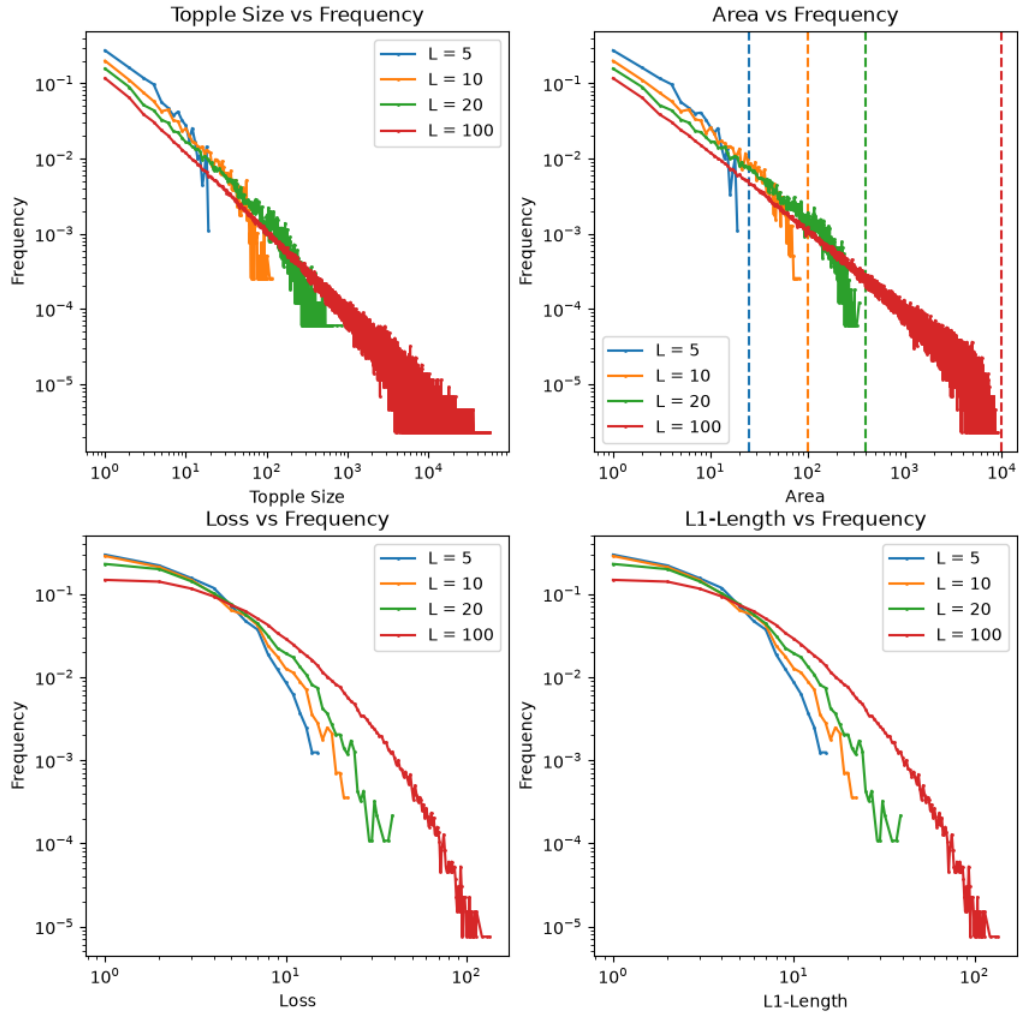


Figure 2.1: Histogram of Topple Size, Area, Loss, and L1 Length on log-log axes, for $L = 5, 10, 20, 100$. Data was generated from $100L^2$ random grain placements, with $4L^2$ burn-in. Black lines in the Area histogram (top-right) are at L^2 and represent the maximum possible toppling area obtainable from an $L \times L$ grid.

Chapter 3

Power Law Fit

1. Naive Power Law Fit
2. Hard Truncated Power Law Fit
3. (Optional) Note on Linear Regression

3.1 Naïve Power Law Fit

We hypothesize a power law scaling relationship for Toppling Size S and area A of the form:

$$p(S = s; \tau, s_{max}) \propto s^{-\tau_s} p(A = a; \tau) \propto a^{-\tau_a}$$

In the above, s_{max} and a_{max} are known constants.

Let $\mathbf{s} = (s_1, \dots, s_T)^T$, $\mathbf{a} = (a_1, \dots, a_T)^T$ be vectors representing the simulation data for topple size and area respectively.

The likelihood function for topple size (and similarly area) is:

$$\mathcal{L}(\mathbf{s}; \tau_s, s_{max}) = \prod_{i=1}^T \frac{s_i^{-\tau_s}}{C_{\tau_s, s_{max}}}$$

$$\log \mathcal{L}(\mathbf{s}; \tau_s, s_{max}) = -T \log C_{\tau_s, s_{max}} - \tau_s \sum_{i=1}^T \log s_i$$

Where $C_{\tau_s, s_{max}}$ is the normalising constant:

$$C_{\tau_s, s_{max}} = \sum_{k=1}^{s_{max}} k^{-\tau_s}$$

To estimate the parameters τ_s and τ_a , we compute their Maximum Likelihood Estimates (MLEs). These are the values $\hat{\tau}_s$ and $\hat{\tau}_a$ that satisfy

$$\hat{\tau}_s = \arg \max_{\tau_s} \mathcal{L}(\mathbf{s}; \tau_s, s_{max}) = \arg \min_{\tau_s} \{-\log \mathcal{L}(\mathbf{s}; \tau_s, s_{max})\}$$

and similarly for $\hat{\tau}_a$.

Because the normalising constant $C_{\tau_s, s_{max}}$ depends on the parameter τ_s , we cannot find an analytic solution and rely on numerical optimisation techniques. The numerical optimisation technique used here is the L-BFGS-B algorithm introduced by Byrd et al. (1995) [2] and implemented in the `scipy` python library.

Initially, we ignore the non-linear tails of the distribution and model the power law as though it is valid for all plausible values. For area, there is a natural upper bound $a_{max} = L^2$. For topple size, which is in theory unbounded, we choose an arbitrary upper bound $s_{max} = 2L^2$ which exceeded all observed topple sizes to compute the normalising constant.

The estimates obtained are presented in Figure 3.1 and Table 3.1.

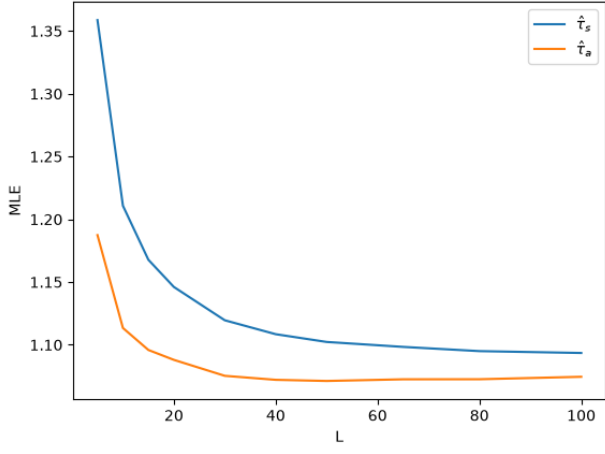


Figure 3.1: Estimates MLEs for τ_s, τ_a for varying lattice sizes L .

L	τ_s	τ_a
5	1.36	1.17
10	1.22	1.12
15	1.17	1.09
20	1.14	1.09
30	1.12	1.07
40	1.11	1.07
50	1.10	1.07
65	1.10	1.07
80	1.09	1.07
100	1.09	1.08

Table 3.1: MLE power-law slopes obtained for varying lattice sizes L

However, inspection of the fit against data in Figure 3.2 indicates that the estimated slope has been overestimated due to the non-powerlaw behaviour of the system at the tails. This effect appears to be most prominent in smaller lattice sizes, but still visibly impacts the fit for $L = 100$.

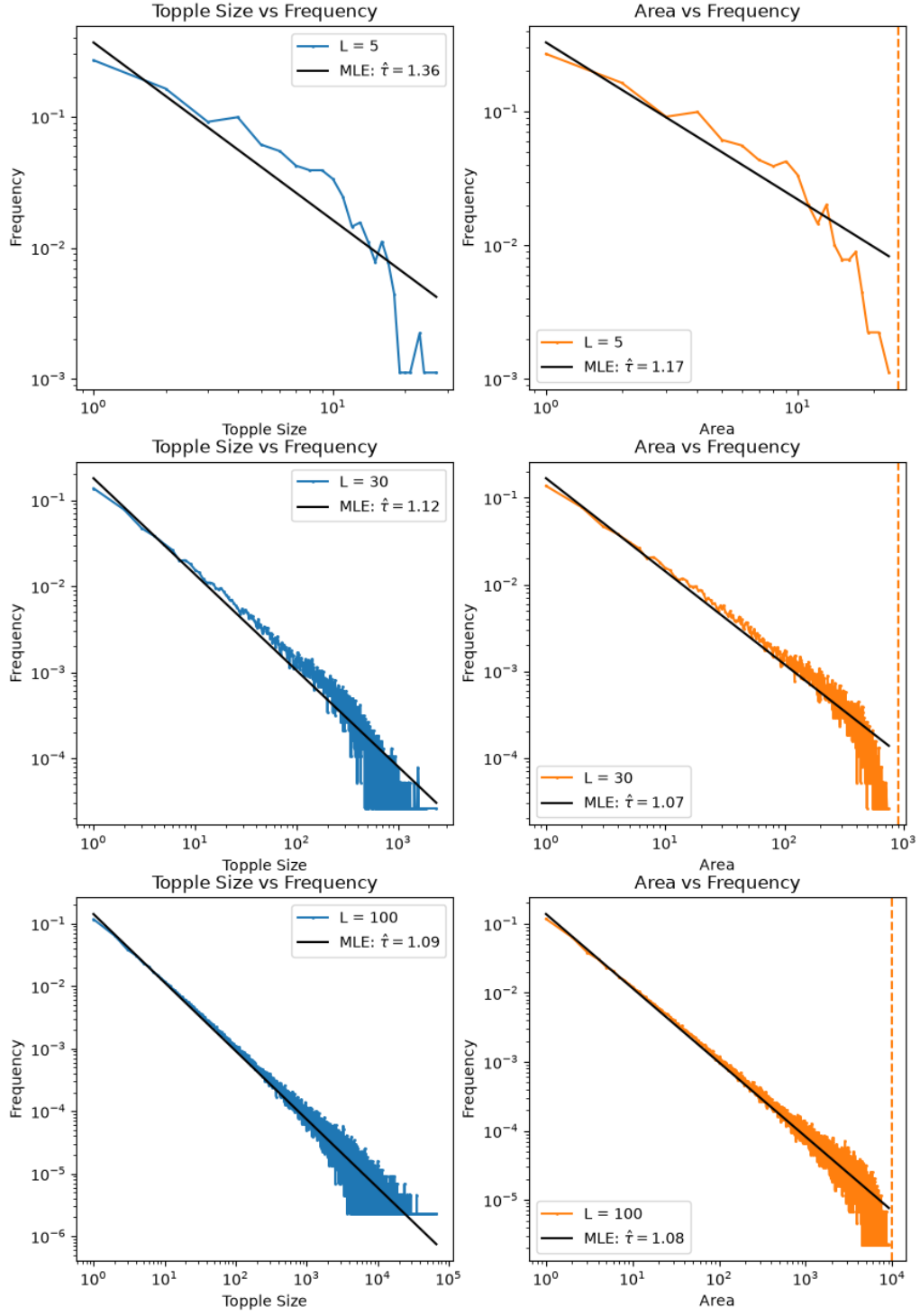


Figure 3.2: Fitted MLEs for a power law distribution against the observed data.

3.2 Manually-truncated Power Law

As a first correction for the finite lattice effect, we consider restricting s_{max} and a_{max} to include only those topple sizes and areas for which linearity is observed in the log-log plot. To avoid manually setting an arbitrary maximum for each value of L , here we consider the rule-of-thumb approximation $s_{max} = a_{max} = 0.1L^2$. Clearly, there are significant limitations to this approach: the cutoff rule was chosen arbitrarily, and data is significantly limited for lower values of L . However this will help us get a more accurate idea of the power law structure. The results in Table 3.2 and Figure 3.3 are noticeably lower than the previous estimates in Table 3.1, with an increasing trend rather than decreasing. From these two analyses, it is reasonable that the powerlaw-governed section of the distribution has exponent $\tau_s, \tau_a \approx 1$, which is consistent with previous papers [1].

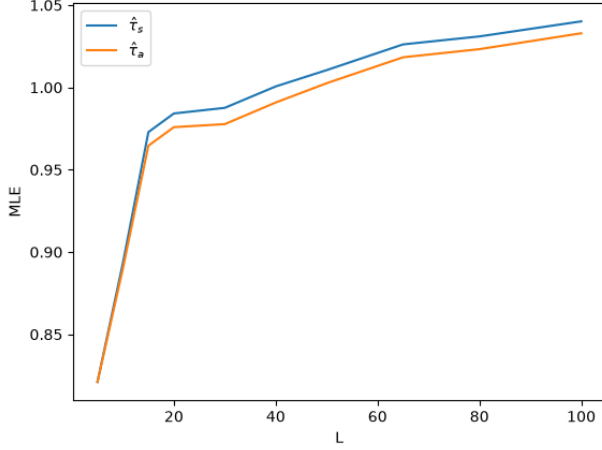


Figure 3.3: Estimates MLEs for τ_s, τ_a for varying lattice sizes L .

L	τ_s	τ_a
5	0.82	0.82
10	0.89	0.89
15	0.97	0.96
20	0.98	0.98
30	0.99	0.98
40	1.00	0.99
50	1.01	1.00
65	1.03	1.02
80	1.03	1.02
100	1.04	1.03

Table 3.2: MLE power-law slopes obtained for varying lattice sizes L under the truncation rule $s_{max} = a_{max} = L^2/10$

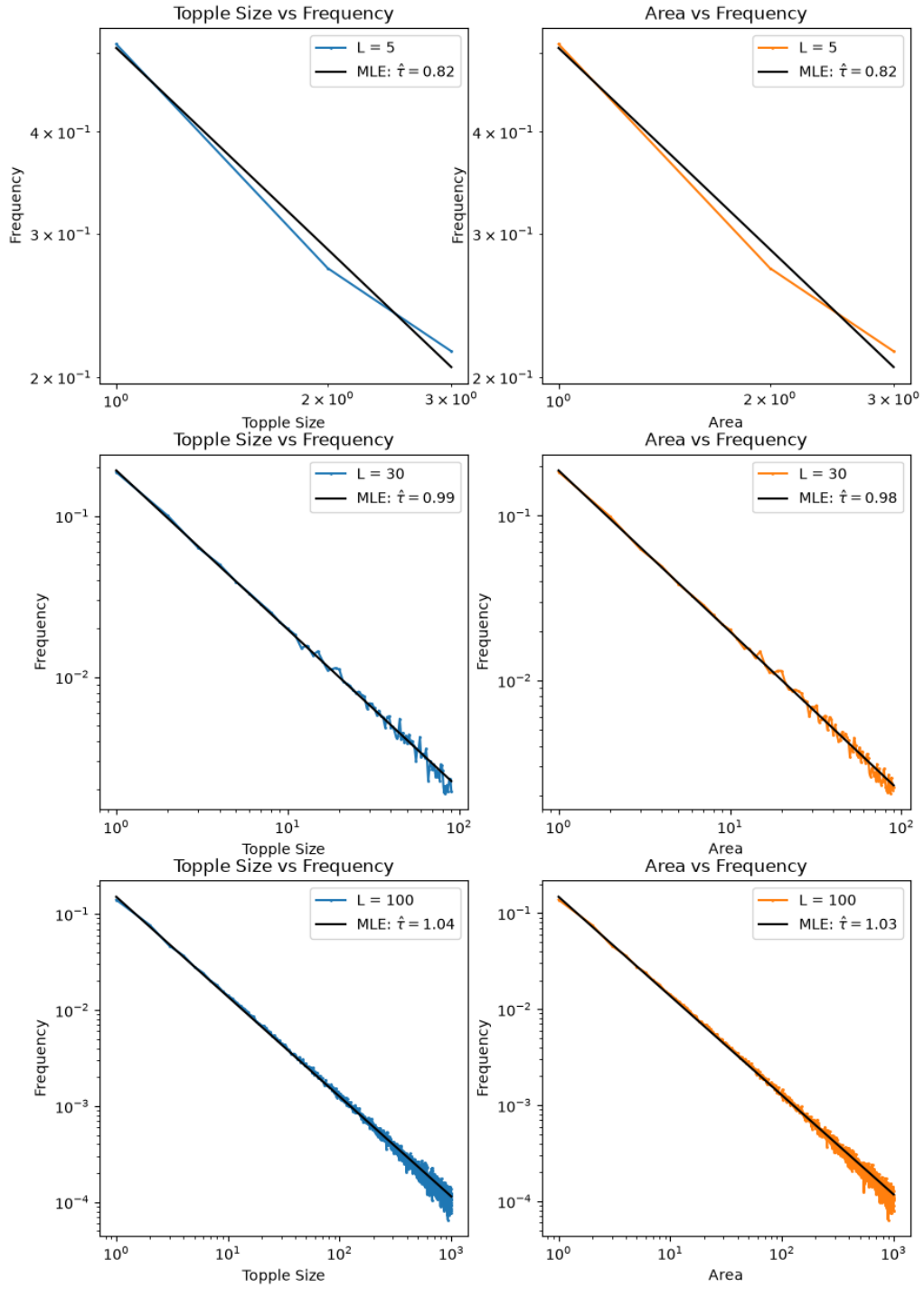


Figure 3.4: Fitted MLEs for a power law distribution against the observed data.

Chapter 4

Finite Size Scaling Effect

1. General form
2. MLE fit for exponentially-truncated power law
3. MLE fit for generalised exp-truncated power law
4. Discussion

The finite size scaling law hypothesizes that the tail behaviour relates to the finite lattice size L via:

$$p(A = a; \tau_a, L, f, D) \propto a^{-\tau_a} f\left(\frac{a}{L^D}\right)$$

Here f governs the tail behaviour of the distribution, i.e. for small values of u , $f(u) \approx 1$ and the distribution resembles a power law.

4.1 Exponentially Truncated Power Law

We first consider the exponentially-truncated power law with $a_{max} = L^2$ as the natural cutoff. Again, we impose $s_{max} = 2L^2$ for computational convenience.

$$p(A = a; \tau_a, c_a, a_{max}) = \frac{1}{C_{\tau_a, c_a, a_{max}}} a^{-\tau_a} \exp\left(\frac{-a}{c_a}\right)$$

Where $C_{\tau_a, c_a, a_{max}}$ is the normalising constant:

$$C_{\tau_a, c_a, a_{max}} = \sum_{k=1}^{a_{max}} k^{-\tau_a} \exp(-k/c_a)$$

The likelihood and log-likelihood are:

$$\begin{aligned} \mathcal{L}(\mathbf{a}; \tau_a, c_a, a_{max}) &= \prod_{i=1}^T \frac{1}{C_{\tau_a, c_a, a_{max}}} a_i^{-\tau_a} \exp\left(\frac{-a_i}{c_a}\right) \\ \log \mathcal{L}(\mathbf{a}; \tau_a, c_a, a_{max}) &= -T \log C_{\tau_a, c_a, a_{max}} - \tau_a \sum_{i=1}^T \log a_i - \frac{1}{c_a} \sum_{i=1}^T a_i \end{aligned}$$

In the above, we model $f(u) = \exp(-u)$. We expect $c_a \propto L^D$ for some constant D across all lattice sizes. The advantage of this structure is that the full (not truncated) dataset can be used to fit the powerlaw exponent τ_a , with c_a representing a soft cutoff boundary that is fit to the data, rather than a hard cutoff that is chosen arbitrarily. If the distribution obeys finite size scaling with this f , then we should expect to see a relatively constant τ_a across varying L , and we should observe a linear $\log(c_a)$ vs $\log(L)$ plot with constant slope D_a . In particular, we can estimate pooled MLEs from the full dataset across different values of L according to

$$p(a; L, \tau_a, \gamma_a, D_a) = \frac{1}{C_{L, \tau_a, \gamma_a, D_a}} a^{-\tau_a} \exp\left(\frac{a}{\gamma_a L^{D_a}}\right)$$

Denote $\mathbf{a}_L$ the set of observations generated from the lattice of size L . Then likelihood and log-likelihood in the pooled case are:

$$\mathcal{L}(\mathbf{a}_{L_1}, \dots, \mathbf{a}_{L_K}; \tau_a, \gamma_a, D_a) = \prod_{j=1}^K \mathcal{L}(\mathbf{a}_{L_j}; \tau_a, c_a = \gamma_a L_j^{D_a}, a_{max} = L^2)$$

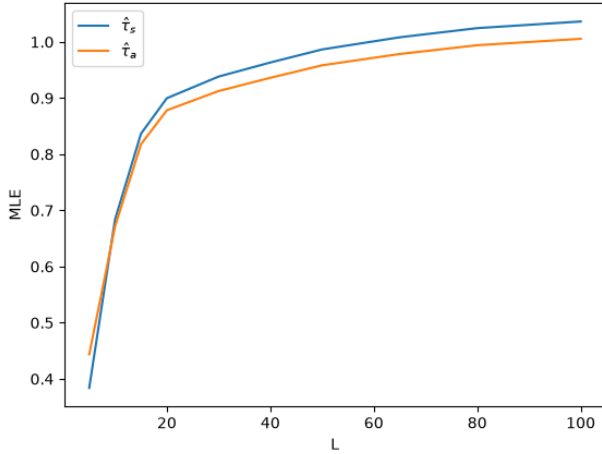
We follow the below procedure:

1. For each L , estimate MLE parameters $\hat{\tau}_a(L), \hat{c}_a(L)$.
2. Examine the MLE fit against the observed data
3. Pool the data and estimate MLE parameters $\tau_a, \gamma_a, \hat{D}_a$.
4. Examine the MLE fit of $\log \hat{\gamma}_a, \hat{D}_a$ against $\log \hat{c}_a(L)$ and $\log L$.

4.1.1 Estimating MLE parameters for each L

As before, for each L , we numerically optimise the log-likelihood function to obtain:

$$\hat{\tau}_a(L), \hat{c}_a(L) = \arg \min_{\tau_a, c_a} \{-\log \mathcal{L}(\mathbf{a}_L; \tau_a, c_a, a_{max} = L^2)\}$$



L	τ_s	c_s	τ_a	c_a
5	0.38	5.69	0.44	6.27
10	0.68	27.49	0.67	26.58
15	0.84	81.55	0.82	74.76
20	0.90	169.69	0.88	148.70
30	0.94	444.13	0.91	355.15
40	0.96	902.92	0.94	671.42
50	0.99	1623.38	0.96	1124.59
65	1.01	3237.96	0.98	2020.08
80	1.02	5846.02	0.99	3306.74
100	1.04	10599.48	1.01	5372.85

Table 4.1: MLE power-law slopes τ and soft cutoffs c obtained for varying lattice sizes L

Figure 4.1: Estimates MLEs for τ_s, τ_a for varying lattice sizes L .

The results in Table 4.1 are promising: for lattice sizes 40 and above, the powerlaw coefficient $\tau \approx 1$. Further, the soft cutoffs c appear to scale appropriately with L , a relationship that we will examine in more detail when computing the pooled estimators.

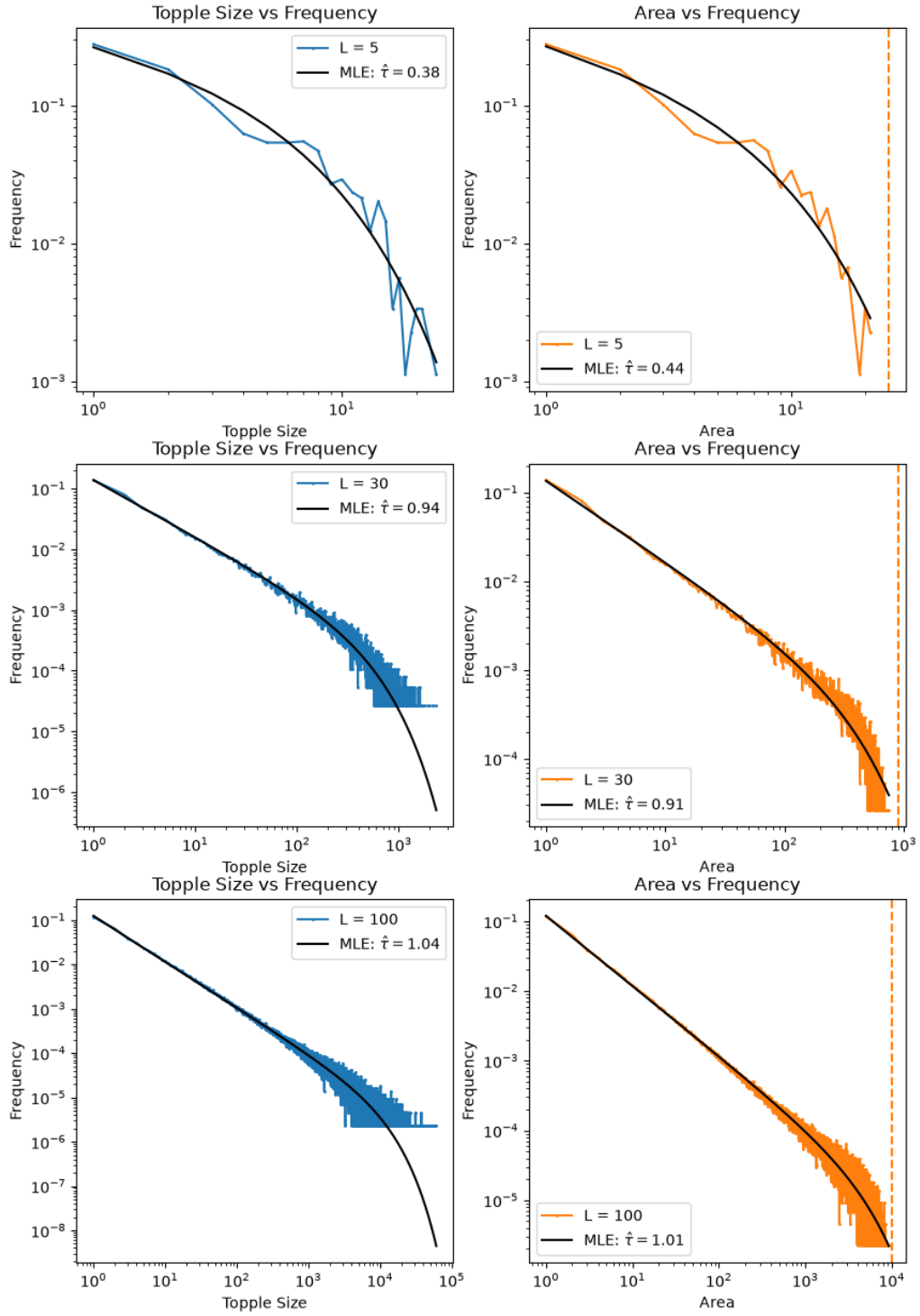


Figure 4.2: Fitted MLEs for the exponentially-truncated power law distribution against the observed data.

Chapter 5

Non-FSS Behaviour of Toppling Number

1. Run-through of previous analysis with the Toppling Number
2. Identify limitations of the analysis

Bibliography

- [1] P. Bak, C. Tang, and K. Wiesenfeld. Self-organized criticality: An explanation of the $1/f$ noise. *Physical review letters*, 59(4):381, 1987.
- [2] R. H. Byrd, P. Lu, J. Nocedal, and C. Zhu. A limited memory algorithm for bound constrained optimization. *SIAM Journal on scientific computing*, 16(5):1190–1208, 1995.